LPMF — Presentation Script

Load Profile Management and Forecasting

Slide-by-slide speaker notes in Hinglish, with formulas, worked examples and viva Q&A

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Poori kahani 1 minute mein

Agar examiner kahe 'ek minute mein batao project kya hai', to yeh bolo (~60 second):

> Delhi ki bijli ki demand har saal garam, peaky aur mehngi hoti ja rahi hai. 29 June 2026 ko Delhi ka peak 8,748 MW tak gaya. DISCOM ko har din agle din ki demand 96 blocks mein, yaani har 15 minute ke liye, batani padti hai, aur galti hone par CERC ke DSM rules ke under paisa lagta hai. Humne ek dashboard banaya jo sirf forecast nahi dikhata, balki uska nateeja bhi dikhata hai. Real Delhi SLDC load data (April 2023 se January 2026) aur Open-Meteo weather ko saaf kiya, 15-minute blocks banaye, aur ek LightGBM model train kiya jo agle din ke 96 blocks predict karta hai. Test period par iski error (MAPE) 3.93% hai, jabki simple seasonal-naive baseline ki 5.16%. Har forecast ke saath ek P10-P90 band bhi hai, jisme 84.4% actual values aayi, target 80% tha. Dashboard ke saare 12 pages live hain: chhe core forecast-pipeline par (Overview, Load Analytics, Weather Impact, Forecast, DSM ₹ Calculator, Transformer Health), aur chhe additional engines jo apne-apne real ya foreign benchmark se check hue hain (IEX Price, ToD Bill Simulator, Solar & Net Load, Losses & PF, Theft & Segments, Data & Model Card). Har number par label hai ki woh REAL hai, SYNTHETIC hai, MODELLED hai, FOREIGN hai ya ILLUSTRATIVE. Aur hum apni limitations bhi khul ke batate hain: Diwali jaise din par error zyada hai, test set mein garmi ka peak season nahi hai, aur Solar Plant 2 ka fit genuinely weak hai, jiska root cause investigate kiya gaya.

Timing table

Speed ka andaaza: ~140 shabd per minute. Viva ke liye 3-5 minute alag rakho. Agar time kam ho, slide 5 (Dataset), slide 6 (Accuracy) aur slide 10 (IEX/ToD) thoda chhota kiya ja sakta hai.

Slide	Title	Time (sec)	Clock (end)
1	Title	35	0:35
2	Problem Statement	75	1:50
3	Gap and USP	70	3:00
4	Dashboard: 12 of 12 Live	55	3:55
5	Dataset & Variables	70	5:05
6	Model Accuracy & Fit Check	75	6:20
7	Overview & Load Analytics	125	8:25
8	Weather Impact & Forecast	135	10:40
9	DSM ₹ Calculator & Transformer Health	120	12:40
10	IEX Price & ToD Bill Simulator	110	14:30
11	Solar & Net Load, and Losses & PF	115	16:25
12	Theft & Segments, and Data & Model Card	105	18:10
13	Benchmarks	50	19:00
14	References	25	19:25
15	Thank You	25	19:50
	Total	**1190**	**19:50 (≈ 19.8 min)**

Slide 1 — Title

Slide par kya hai

Project ka naam (Load Profile Management & Forecasting), Faculty of Technology, University of Delhi, team ke chaar naam aur dono guides: Dr. Ujjal Sur aur Dr. Arjun Tyagi.

Bolne wali script (~35 sec)

Good morning sir/ma'am. Hum aaj apna project present kar rahe hain: Load Profile Management and Forecasting, yaani Delhi grid ke liye ek AI-integrated load intelligence dashboard. Main hoon Vansh Sachan, aur mere saath hain Animish Gupta, Ayush Rathore aur Lavish Rawat. Yeh project humne Dr. Ujjal Sur aur Dr. Arjun Tyagi ki guidance mein kiya hai. Agle kuch minute mein hum batayenge ki problem kya hai, humne kya banaya, aur har number kahan se aaya.

Kya hua / Kyu kiya / Kaise kiya

- Kya hua: Project aur team ka introduction.
- Kyu kiya: Audience ko pehle hi pata chal jaye ki topic power system aur machine learning dono ka hai.
- Kaise kiya: Ek line ka subtitle: 'AI-integrated load intelligence dashboard for the Delhi grid'.

Viva mein pooche ja sakte hain

Q: Project ka one-line summary kya hai?

A: Real Delhi data se agle din ki 96-block load forecast, aur us forecast ko peak alert, DSM ₹ aur transformer ageing jaise engineering nateejon tak le jaana.

Q: Isme AI kahan hai?

A: Forecast LightGBM model se aata hai, jo ek machine learning model hai. Baaki pages engineering formulas hain, except Theft & Segments jahan dusra ML model (LightGBM classifier) hai.

Agli slide par: To chaliye, pehle dekhte hain ki Delhi ki demand ki problem itni serious kyun hai.

Slide 2 — Problem Statement

Slide par kya hai

Chaar bade numbers: India ka all-time peak 256.1 GW, Delhi ka 8,748 MW, heat index se 67% daily-peak variation (CSE 2025), aur IEX price ₹1.11 se ₹9.71 per unit. Niche problem statement aur kaun affected hai.

Bolne wali script (~75 sec)

Delhi ki demand garam, peaky aur galat hone par mehngi hai. 25 April 2026 ko India ka peak 256.1 GW tha, aur 29 June 2026 ko Delhi ka record 8,748 MW. CSE ki 2025 study kehti hai ki Delhi ke daily peak ke variation ka lagbhag 67% heat index se samjha ja sakta hai. Heat index matlab temperature aur humidity ko milake 'kitni garmi mehsoos hoti hai'. DISCOMs ko agle din ki demand 96 blocks of 15 minutes mein batani padti hai. Agar forecast galat ho, to CERC DSM Regulations 2024 ke under har MW deviation ka charge lagta hai. Saath hi, 40 degree ke din full load par transformer ki insulation lagbhag 10 guna tez umar kha jaati hai, solar ki wajah se shaam ki demand tezi se chadhti hai, aur AT&C losses abhi bhi 15.04% hain. Aaj ke monitoring tools sirf bata dete hain ki kya ho chuka; yeh nahi batate ki kal ki galti ka kya nateeja hoga.

Kya hua / Kyu kiya / Kaise kiya

- Kya hua: Problem ko numbers ke saath define kiya.
- Kyu kiya: Sirf 'forecast karna hai' kehne se importance samajh nahi aati; paisa, equipment aur grid par asar dikhana zaroori tha.
- Kaise kiya: Har number ke saath official source likha: PIB, Delhi SLDC, CSE, EAC-PM, IEC 60076-7, 14th Integrated Rating.

Formula / number ka matlab

96 blocks ka matlab

1 din = 24 ghante × 4 blocks per ghanta = 96 blocks, har block 15 minute ka. DSM settlement isi block level par hota hai, isliye forecast bhi 96 blocks mein diya jaata hai.

Viva mein pooche ja sakte hain

Q: 67% wala number aapka hai?

A: Nahi, yeh CSE 2025 ka hai aur daily peak ke liye hai. Hamara apna fit har 15-min block par r^2 = 0.61 deta hai. Dono alag metric hain, isliye direct compare nahi kar sakte.

Q: 8,748 MW aapke data mein hai?

A: Nahi. Hamara data 12 January 2026 par khatam hota hai, aur yeh peak 29 June 2026 ka hai. Data ka maximum 8,581.6 MW (19 June 2024) hai. Isliye 8,748 MW sirf problem-statement context ke liye hai.

Agli slide par: Ab sawal yeh hai ki baaki projects yeh problem kyun solve nahi karte; yeh agli slide mein.

Slide 3 — Gap and USP

Slide par kya hai

Do column: typical load-forecasting projects kya karte hain, aur hamara system kya alag karta hai. Niche USP box.

Bolne wali script (~70 sec)

Zyadatar forecasting projects ek graph dikha ke ruk jaate hain, jaise 'kal 8,000 MW'. Ek model, ek accuracy number, aur aksar foreign ya toy dataset. Hum forecast ko uske nateeje tak le jaate hain. Pehla, model ladder: har complex model ko apne se simple model ko beat karna padta hai, warna hum use nahi rakhte. Aur hum imaandari se batate hain ki h = 1-2, yaani 15-30 minute aage, simple persistence abhi bhi jeet jaata hai. Doosra, data real hai: Delhi SLDC ka 5-minute data aur Open-Meteo weather. Teesra, har forecast ke saath P10-P90 band hai, aur selected din par SHAP explanation. Chautha, DSM ₹ aur transformer life ke what-if calculators. Hamara USP: design ke hisaab se ek Delhi forecast chhe engines chalata hai, aur ab ToD bill, solar ramp, losses aur theft bhi live hain - sirf forecast nahi, har engine ship ho chuka hai. Alag-alag tools GitHub aur Kaggle par milte hain, lekin unhe Indian regulation par ek saath jodna naya hai.

Kya hua / Kyu kiya / Kaise kiya

- Kya hua: Existing kaam aur hamare kaam ka farak saaf kiya.
- Kyu kiya: Examiner ka pehla sawal hota hai 'isme naya kya hai?'.
- Kaise kiya: Paanch point side-by-side compare kiye; sabhi 12 engines ab live hain, koi future scope baaki nahi.

Viva mein pooche ja sakte hain

Q: Persistence kya hai?

A: Sabse simple forecast: agla block = abhi wala block. 15 minute mein load bahut kam badalta hai, isliye bahut chhote horizon par yeh bahut accurate hota hai.

Q: SHAP kya hai?

A: Ek method jo batata hai ki kis feature ne forecast ko kitna upar ya neeche dhakela. Jaise 'kal ka load' sabse bada driver nikla.

Agli slide par: Ab dekhte hain ki yeh sab dashboard par kaise dikhta hai.

Slide 4 — Dashboard: 12 of 12 Live

Slide par kya hai

12 cards, sab rangeen aur 'LIVE'. Pehle chhe (Overview, Load Analytics, Weather Impact, Forecast, DSM ₹ Calculator, Transformer Health) core forecast-pipeline par hain; agle chhe (IEX Price, ToD Bill Simulator, Solar & Net Load, Losses & PF, Theft & Segments, Data & Model Card) apne-apne benchmark ke saath additional engines hain.

Bolne wali script (~55 sec)

Dashboard mein 12 pages hain, aur har page ek operational sawal ka jawab deta hai. Saare baarah live hain aur real API se chalte hain. Overview batata hai kal ka peak kitna aur kab hoga. Load Analytics batata hai demand kitni peaky hai. Weather Impact batata hai garmi se kitna load badhta hai. Forecast mein 96 blocks, uncertainty band aur model comparison hai. DSM ₹ Calculator aur Transformer Health what-if calculators hain. Baaki chhe pages ab bhi live hain: IEX Price real historical exchange price use karta hai, ToD Bill Simulator real DERC rates, Solar & Net Load real plant data se validate hua, Losses & PF real PFC AT&C figures ke saath, Theft & Segments real SGCC benchmark par, aur Data & Model Card har page ka summary deta hai.

Kya hua / Kyu kiya / Kaise kiya

- Kya hua: Poore system ka map ek slide par.
- Kyu kiya: Sabhi engines ko live dikhana imaandaari hai, lekin har engine ka apna benchmark bhi batana zaroori hai.
- Kaise kiya: Next.js frontend, FastAPI backend; har card provenance badge (REAL, SYNTHETIC, ILLUSTRATIVE, MODELLED, FOREIGN) ke saath.

Viva mein pooche ja sakte hain

Q: Data & Model Card ka data kahan se aata hai?

A: Yeh ek static summary page hai, docs/MODEL_RESULTS.md aur API responses se banaya gaya hai -- koi naya data nahi, sirf har page ka source aur result ek jagah.

Q: Live pages ka data kahan se aata hai?

A: FastAPI se. Har response mein provenance (REAL / ILLUSTRATIVE / MODELLED / FOREIGN / SYNTHETIC) aur source string hoti hai.

Agli slide par: Har page ka number kahan se aata hai, yeh agli slide batati hai: dataset.

Slide 5 — Dataset & Variables

Slide par kya hai

Do tables side-by-side, 12 rows total -- har ek baarah features ka apna source, key variables (exact field names), tag (REAL / MODELLED / ILLUSTRATIVE / FOREIGN-REAL) aur cleaning/validation ka kaaran. Niche note box: 11 of 12 features real ya validated open data par hain.

Bolne wali script (~70 sec)

Yeh slide batati hai ki har number kahan se aata hai, ab saari baarah features ke liye. Pehle chaar, yaani Overview, Load Analytics, Weather Impact aur Forecast, poori tarah REAL data par chalte hain: Delhi SLDC ka load, April 2023 se January 2026, aur Open-Meteo weather Delhi ke coordinates par. Variables wahi hain jo code mein hain, jaise load_mw, heat_index_c, temp_c aur rh_pct. DSM ₹ Calculator aur Transformer Health data-driven nahi hain; user value deta hai aur formula chalta hai, isliye unpar hamesha ILLUSTRATIVE likha hai. Baaki chhe engines bhi apne-apne real benchmark par hain: IEX Price Kaggle ke real IEX day-ahead market data (2018-2024) par, ToD Bill Simulator DERC ke real tariff order par ek illustrative consumer profile ke saath, Solar & Net Load real Open-Meteo GHI par jo do real plants se validate hua, Losses & PF PFC ke real AT&C benchmark ke saath illustrative PF what-ifs, Theft & Segments real SGCC (China) benchmark par -- sirf demo customer-fleet synthetic hai -- aur Data & Model Card ek static real summary hai. Cleaning causal hai, yaani har decision sirf past data dekh kar hota hai: Hampel outlier filter, gap filling, aur train, validation aur test ka chronological split. Overall, baarah mein se gyarah features real ya validated open data par hain, sirf Theft & Segments ka demo fleet synthetic hai.

Kya hua / Kyu kiya / Kaise kiya

- Kya hua: Har feature ka source, variables aur tag do tables mein.
- Kyu kiya: Engineer ko pata hona chahiye ki kis number par kitna bharosa karein; real aur illustrative ko mix karna khatarnaak hai.
- Kaise kiya: Field names seedhe app/schemas.py aur engine functions se liye; har chart par provenance badge.

Formula / number ka matlab

Row counts (DATA_QUALITY.md, data files)

5-minute grid par 293,184 slots chahiye the, 285,252 mile (2.705% missing). 15-minute par 97,728 blocks, jinme se 1,330 (1.361%) khaali chhode gaye, kyunki gap bahut bada tha.

Viva mein pooche ja sakte hain

Q: Causal cleaning kyun zaroori hai?

A: Wahi load series features aur targets dono banati hai. Agar cleaning future dekhe, to future ki jaankari features mein leak ho jaati hai aur score galat lagega.

Q: Gap filling kaise kiya?

A: Gap ki lambai ke hisaab se: chhote gap forward-fill (894 blocks), medium gap pichhle 5 din ke same time ke profile se (1,153 blocks), aur 3 ghante se bade gap khaali chhode (1,330 blocks).

Agli slide par: Model kitna accurate hai, aur kya woh overfit ya underfit hai? Yeh agli slide batati hai.

Slide 6 — Model Accuracy & Fit Check

Slide par kya hai

Left: train / validation / test MAPE ka chart (model vs seasonal-naive) aur teen accuracy tiles. Right: MAPE, accuracy, naive MAPE aur gain ki table; horizon-wise error; overfitting/underfitting verdict aur do honest caveats.

Bolne wali script (~75 sec)

Ab dekhte hain ki model kitna accurate hai, aur kya woh overfit ya underfit hai. Accuracy hum 100 minus MAPE se define karte hain. Test par MAPE 3.93% hai, yaani accuracy lagbhag 96%. Train par MAPE 2.29%, accuracy 97.71%; validation par 4.29%, accuracy 95.71%. Seasonal-naive, yaani kal isi block ka load, train par 5.44%, validation par 5.97% aur test par 5.16% deta hai, isliye model test par 24% kam galti karta hai. Overfitting check: train aur test ke beech gap hai, lekin chhota hai. Early stopping 3,000 mein se sirf 436 rounds par ruki, aur validation aur test error flat rahi; sirf train error giri. Isliye overfitting nuksaan wali nahi hai. Underfitting bhi nahi hai, kyunki model har horizon par naive se behtar hai: h = 1-4 par 3.21%, h = 5-24 par 3.48%, h = 25-48 par 3.84% aur h = 49-96 par 4.22%, jabki naive lagbhag 5.16%. Do honest caveats: pehla, test set poori tarah untouched nahi tha, kyunki target-aligned lags pehle model ki test errors dekh kar design hue, isliye 3.93% thoda optimistic ho sakta hai. Doosra, sabse chhote horizon par persistence model se behtar hai: h = 1 par 1.44% vs 3.18%.

Kya hua / Kyu kiya / Kaise kiya

- Kya hua: Saved model ka train, validation aur test par accuracy aur overfitting / underfitting check.
- Kyu kiya: Sirf ek accuracy number kaafi nahi; examiner poochhega ki model ratta to nahi maar gaya (overfit) ya bahut simple to nahi (underfit).
- Kaise kiya: Saved model ko bina retrain kiye har split ke saare origin × horizon rows par score kiya; seasonal-naive se compare kiya; learning curve aur early stopping check kiya.

Formula / number ka matlab

Accuracy = 100 − MAPE; Gain = 1 − MAPE(model) / MAPE(naive)

Test: 100 − 3.93 = 96.07% ≈ 96%. Train 97.71%, validation 95.71%. Gain: 1 − 3.93 / 5.16 = 24% (validation 28%, train 58%). Early stopping: 436 rounds, limit 3,000 (patience 50).

Viva mein pooche ja sakte hain

Q: Overfitting hai ya nahi?

A: Train-test gap chhota hai (2.29% vs 3.93%), early stopping 436/3000 par ruki, aur validation aur test error flat hain, sirf train girti hai. Isliye overfitting nuksaan wali nahi hai.

Q: 96% accuracy kaise nikali?

A: Accuracy = 100 − MAPE. Test MAPE 3.93% hai, to 100 − 3.93 = 96.07% ≈ 96%. Yeh classification accuracy nahi hai; har block ki galti ko actual load se normalise karke average kiya gaya hai.

Q: Validation test se zyada kyun?

A: Validation ka do-tihai hissa March-June ki garmi hai, aur naive bhi wahan zyada galti karta hai (5.97% vs 5.16%); isliye woh window mushkil hai.

Agli slide par: Ab ek-ek karke saari 12 features dekhte hain, shuru karte hain Overview aur Load Analytics se.

Slide 7 — Overview & Load Analytics

Slide par kya hai

Ek page mein dono features: Overview (peak, peak time, alert) aur Load Analytics (load factor, PAR, duration curve). Dono REAL.

01 · Overview

Tag: REAL

Bolne wali script (~60 sec)

Overview page sabse pehla sawal answer karta hai: kal ka peak kitna hoga, kab hoga, aur kya woh limit cross karega. Formula simple hai: 96 forecast blocks mein sabse bada value peak hai, aur agar woh limit se zyada hai to alert. Hamare test ke aakhri din, 12 January 2026 ka example: sabse bada forecast block 42 mein aaya. Block 42 ka start time hai 41 guna 15 minute, yaani 615 minute, matlab subah 10:15. Peak 5,670.83 MW hai. Limit humne Delhi ka record 8,748 MW rakhi. 5,670.83, 8,748 se kam hai, isliye alert 'Clear' hai, aur lagbhag 3,077 MW ka headroom bacha hai.

Kya hua / Kyu kiya / Kaise kiya

- Kya hua: Agle din ka peak, uska time, aur overload alert.
- Kyu kiya: Operator ko limit cross hone se pehle warning milni chahiye, baad mein nahi.
- Kaise kiya: LightGBM ke 96-block forecast par peak_and_alert engine chalta hai (ml/src/lpmf/engines/overview.py).

Formula / number ka matlab

$\hat{P}peak = \max_t \hat{P}(t)$; alert agar $\hat{P}peak > Plimit$

Viva mein pooche ja sakte hain

Q: Limit 8,748 MW hi kyun?

A: Yeh Delhi ka all-time record hai, isliye ek natural upper limit hai. Code mein limit ek parameter hai, DISCOM apni limit daal sakta hai.

Q: Do blocks ka value same ho to?

A: Engine sabse pehla (earliest) block chunta hai.

02 · Load Analytics

Tag: REAL

Bolne wali script (~65 sec)

Load factor, yaani LF, matlab average load ko peak load se divide karna. PAR, yaani peak-to-average ratio, iska ulta hai. LF kam ho to iska matlab hai ki grid ki capacity zyadatar time khaali padi rehti hai. Pehla illustrative din: average 530, peak 820, to LF = 530 / 820 = 0.646, PAR = 1.55 -- sirf formula samjhane ke liye. Doosra real, poore period ka measured: April 2023 se January 2026 ke 96,398 blocks. Average 4,316 MW, peak 8,582 MW, to LF = 0.503, PAR = 1.99. Graph load-duration curve hai: saare blocks bade se chhote load tak sort kiye.

Kya hua / Kyu kiya / Kaise kiya

- Kya hua: Demand kitni peaky hai, LF, PAR aur duration curve se.
- Kyu kiya: PAR wahi cheez hai jise demand response ko kam karna hota hai; isse capacity planning hoti hai.
- Kaise kiya: load_factor_and_par engine; missing blocks skip hote hain, zero nahi maane jaate.

Formula / number ka matlab

LF = Pavg / Pmax; PAR = 1 / LF

Viva mein pooche ja sakte hain

Q: 0.646 aur 0.503 mein kaunsa sahi hai?

A: Dono sahi hain, lekin alag cheez hain. 0.646 ek illustrative din ka example hai; 0.503 poore data ka measured value hai. Dashboard 0.503 dikhata hai.

Q: Poore period ka LF ek din se kam kyun hai?

A: Lambe period mein peak saal ka sabse bada din hota hai, jabki average saare dino ka. Isliye ratio chhota aata hai.

Agli slide par: Ab dekhte hain is peakiness ka sabse bada kaaran, garmi, aur phir project ka core: Forecast page.

Slide 8 — Weather Impact & Forecast

Slide par kya hai

Ek page mein dono features: Weather Impact (heat index vs load, OLS fit) aur Forecast (96-block forecast, P10-P90 band, SHAP). Dono REAL.

03 · Weather Impact

Tag: REAL

Bolne wali script (~65 sec)

Yeh page batata hai ki har ek degree heat index badhne par load kitna badhta hai. Humne ordinary least squares, yaani OLS, se ek seedhi line fit ki, 96,398 fifteen-minute blocks par. Result: β = 112.0 MW per degree Celsius. Iska matlab heat index 3 degree badhe to load lagbhag 112 x 3 = 336 MW badhta hai. r^2 = 0.61, matlab load ke variation ka 61% heat index se explain hota hai. Dhyaan dijiye, CSE ka 67% alag hai: woh daily peak par hai, hamara har 15-minute block par.

Kya hua / Kyu kiya / Kaise kiya

- Kya hua: Garmi se load kitna badhta hai, MW per °C mein.
- Kyu kiya: Heat sabse bada driver hai; β se pata chalta hai ki garam din ke liye kitna margin chahiye.
- Kaise kiya: heat_sensitivity engine har baar data se β dobara nikalta hai; heat index Rothfusz formula se, temp_c aur rh_pct se.

Formula / number ka matlab

$P = \alpha + \beta \cdot HI$; $\Delta P \approx \beta \cdot \Delta HI$; $HI = f(T, RH)$

Viva mein pooche ja sakte hain

Q: Graph mein curve U-shape jaisi kyun hai, line seedhi kyun?

A: Thandi mein heating ki wajah se load thoda badhta hai, isliye neeche U jaisa dikhta hai. Humne ek simple linear fit rakha; isliye r^2 0.61 hai, 1 nahi.

Q: Scatter daily means ka hai, fit blocks par kyun?

A: 96,398 points plot karne se graph bhar jaata, isliye 1,015 daily points dikhaye. β aur r^2 sab blocks par nikle hain.

04 · Forecast

Tag: REAL

Bolne wali script (~70 sec)

Yeh project ka dil hai. LightGBM ek hi model se agle din ke saare 96 blocks predict karta hai. Poore test set, 13 July 2025 se 12 January 2026, par MAPE 3.93% hai, seasonal-naive ka 5.16%, error lagbhag 24% kam hua. Har forecast ke saath P10-P90 band hai, ek 80% range. Target tha 80% actual values band ke andar aayein; test par 84.4% aayi. SHAP bars batate hain ki forecast kyun aaya.

Kya hua / Kyu kiya / Kaise kiya

- Kya hua: Agle din ke 96 blocks ka forecast, uncertainty band ke saath.
- Kyu kiya: DSM charge forecast error par lagta hai; isliye accurate forecast aur imaandaar uncertainty dono chahiye.
- Kaise kiya: Multi-horizon LightGBM (37 features, Optuna se tune, validation par early stopping), MAPIE split-conformal band, TreeSHAP.

Formula / number ka matlab

$\text{MAPE} = (100/n)\ \Sigma\ |y - \hat{y}| / y$; $\text{Coverage} = N(P10 \le y \le P90) / n$

Viva mein pooche ja sakte hain

Q: Error horizon ke saath kaise badalta hai?

A: h = 1-4 par 3.21%, h = 5-24 par 3.48%, h = 25-48 par 3.84% aur h = 49-96 par 4.22%. Seasonal-naive har horizon par lagbhag 5.16% rehta hai.

Q: Sabse important feature kaunsa nikla?

A: tlag_1day, yaani kal isi block ka load: 66.5% gain importance. Ise shuffle karne par MAPE 3.93% se 23.4% ho jaata hai, matlab yeh genuine signal hai, leak nahi.

Agli slide par: Forecast mein galti hogi hi; agla page batata hai ki us galti ki keemat kya ho sakti hai, aur equipment par kya asar padta hai.

Slide 9 — DSM ₹ Calculator & Transformer Health

Slide par kya hai

Ek page mein dono features: DSM ₹ Calculator (deviation → rupees) aur Transformer Health (hot-spot temp → ageing). Dono ILLUSTRATIVE.

05 · DSM ₹ Calculator

Tag: ILLUSTRATIVE

Bolne wali script (~60 sec)

DSM ka matlab Deviation Settlement Mechanism: agar DISCOM schedule se zyada ya kam bijli le, to charge lagta hai. Hamara formula: volume limit tak ka deviation normal rate NR par, aur limit se upar wala hissa k guna rate par. Example: deviation 60 MW, limit 41 MW, NR ₹5,000 per MWh, k = 1.5. Limit tak: ₹51,250; upar: ₹35,625. Total ₹86,875 per block, 4 blocks mein ₹3,47,500 = ₹3.475 lakh/hr.

Kya hua / Kyu kiya / Kaise kiya

- Kya hua: Forecast deviation ko rupees mein badalna.
- Kyu kiya: Kuch MW ki galti abstract lagti hai; rupees mein dikhane par uski keemat samajh aati hai.
- Kaise kiya: dsm_penalty engine; over- aur under-drawal dono |D| se charge; output hamesha is_illustrative = True.

Formula / number ka matlab

$Cb = \min(|D|, VL) \cdot NR/4 + \max(|D| - VL, 0) \cdot k \cdot NR/4$

Viva mein pooche ja sakte hain

Q: Yeh number real penalty kyun nahi?

A: CERC 2024 mein normal rate grid frequency se judi hoti hai; woh tables encode nahi kiye. Humne ek flat rate aur ek multiplier maana hai, isliye sirf illustrative hai.

Q: 41 MW limit kahan se?

A: Yeh BRPL ke liye DERC 2024 intra-state DSM order wali volume limit ke roop mein li gayi hai; NR = ₹5,000 assumed hai. Calculator mein user limit ko slider se badal sakta hai.

06 · Transformer Health

Tag: ILLUSTRATIVE

Bolne wali script (~60 sec)

Transformer ki insulation garmi se budhi hoti hai. IEC 60076-7 standard ke hisaab se 98 degree hot-spot par ageing rate 1 hai; har 6 degree badhne par yeh double ho jaata hai. Example: 40 degree ambient aur full load par hot-spot = 40 + 55 + 23 = 118 degree. Tab V = 2^((118 − 98)/6) = 2^3.33 = 10.08. Matlab 118 degree par ek ghanta chalane se insulation ki 10.08 ghante ki life khatam hoti hai.

Kya hua / Kyu kiya / Kaise kiya

- Kya hua: Hot-spot temperature se insulation ageing aur life used.
- Kyu kiya: Garam dino mein kaunsa ghanta transformer ko sabse zyada nuksaan karta hai, yeh dikhana.
- Kaise kiya: transformer_ageing engine; 118 °C default steady_state_hotspot_c se (Δθor = 55 K, H·gr = 23 K, K = 1).

Formula / number ka matlab

$V = 2^{((\theta h - 98)/6)}$; $L = \Sigma\, Vn \cdot \Delta t$

Viva mein pooche ja sakte hain

Q: 55 aur 23 kahan se aaye?

A: IEC 60076-7 ke distribution transformer (ONAN) defaults: top-oil rise 55 K aur hot-spot gradient 23 K. Rated load par yeh ambient se 78 K upar le jaate hain.

Q: Yeh page real kyun nahi?

A: Hamare paas kisi transformer ka temperature data nahi hai; user temperature deta hai aur formula chalta hai.

Agli slide par: Yeh the humare original chhe forecast-pipeline pages. Ab dekhte hain woh chhe engines jo isi forecast par apne benchmark ke saath bane hain, shuru karte hain IEX Price se.

Slide 10 — IEX Price & ToD Bill Simulator

Slide par kya hai

Ek page mein dono features: IEX Price (real historical block-of-day price) aur ToD Bill Simulator (real DERC tariff). Dono REAL.

07 · IEX Price

Tag: REAL

Bolne wali script (~55 sec)

Humne Kaggle se real Indian Energy Exchange ka day-ahead market data liya, 2018 se 2024 tak, 233,760 rows. Har block-of-day (din ke 96 slots) ka average price nikala aur usse day-ahead load ke saath multiply karke purchase cost banaya, purani illustrative cosine curve ki jagah. Yeh real historical price range ₹3.05 se ₹6.13 per unit deta hai, jo exchange ki apni published range se match karta hai.

Kya hua / Kyu kiya / Kaise kiya

- Kya hua: Forecast ko real historical market price se jodna.
- Kyu kiya: Sirf load forecast kaafi nahi; DISCOM ko yeh bhi pata hona chahiye ki woh load kitne mein kharidega.
- Kaise kiya: Real historical block-of-day average (ek specific din ka live price nahi); Kaggle mukund604/indian-energy-exchange-time-series-forecasting dataset se.

Formula / number ka matlab

cost = Σ P_load(t) · 0.25h · price(t)

Viva mein pooche ja sakte hain

Q: IEX price kaise verify kiya?

A: Humara number IEX ki apni published range ke saath compare kiya -- match karta hai, lekin yeh ek 6.7-saal ka typical-day average hai, kisi specific din ka live price nahi.

Q: Purani illustrative curve ka kya hua?

A: Replace ho gayi real historical average se; agar pipeline nahi chala to API uss illustrative curve par fall back karta hai, reported nahi hota.

08 · ToD Bill Simulator

Tag: REAL + ILLUSTRATIVE

Bolne wali script (~55 sec)

DERC ka Tariff Schedule FY2020-21, clause 3(d), seedha PDF se padha: peak hours (14:00-17:00 aur 22:00-01:00) mein 20% surcharge, off-peak (04:00-10:00) mein 20% rebate, sirf May se September. Flat bill vs ToD bill vs load shift karne ke baad ka bill, teeno dikhaye jaate hain -- tariff real hai, consumer load profile illustrative what-if hai.

Kya hua / Kyu kiya / Kaise kiya

- Kya hua: Ek illustrative load profile par real DERC tariff lagana.
- Kyu kiya: Consumer ko dikhana ki peak-hour load shift karne se bill kitna bach sakta hai.
- Kaise kiya: DERC ka primary PDF directly fetch karke parse kiya, pehle ek unverified figure correct karte hue.

Formula / number ka matlab

bill = Σ(h=0 to 23) kWh(h) · rate · m(h), jahan m(h) peak/off-peak multiplier hai

Viva mein pooche ja sakte hain

Q: ToD rates DERC se seedhe aaye?

A: Haan. Pehle ek unverified 30%/25% figure use kiya tha; fir directly DERC ke FY2020-21 tariff schedule PDF se clause 3(d) padh kar 20%/20% correct kiya.

Q: Yeh schedule abhi bhi valid hai?

A: Haan, DERC ne 30 Sep 2021 ke baad koi naya base tariff order nahi diya; baad ke orders sirf fuel-cost surcharge adjust karte hain, ToD structure nahi.

Agli slide par: Ab solar aur losses dekhte hain -- dono mein real plant aur real regulator data use hua hai.

Slide 11 — Solar & Net Load, and Losses & PF

Slide par kya hai

Ek page mein dono features: Solar & Net Load (PVWatts-style formula, real-plant validated) aur Losses & PF (real AT&C benchmark + illustrative PF what-ifs).

09 · Solar & Net Load

Tag: MODELLED (real-plant validated)

Bolne wali script (~60 sec)

Formula simple hai: GHI (irradiance) ko capacity aur performance ratio se multiply karo. Isse do real Indian solar plants ke generation data (Kaggle) se bhi validate kiya. Plant 1 par formula bahut achha fit hua, daytime r^2 = 0.97. Plant 2 par daytime r^2 sirf 0.21 nikla -- humne chaar cheezein check ki: timestamp shift, inverter clipping, sensor placement aur din-wise pattern. Koi fixable bug nahi mila; asli wajah hai ek hi weather sensor jo 22 alag-alag inverters ke liye use hota hai, aur Nasik ke monsoon onset ke time cloud variability badh jaati hai.

Kya hua / Kyu kiya / Kaise kiya

- Kya hua: Solar output ko real plant data se validate karna.
- Kyu kiya: Formula ko blindly trust nahi karte -- real data par test karte hain, aur weak fit ko chhupane ke bajaye root-cause investigate karte hain.
- Kaise kiya: Solar formula ko do real plants ke CSV (Kaggle) ke against validate kiya.

Formula / number ka matlab

$P_{solar}(t) = (GHI(t)/1000) \cdot C_{kWp} \cdot PR$; $P_{net} = P_{load} - P_{solar}$

Viva mein pooche ja sakte hain

Q: Solar Plant 2 ka r^2 kam kyun hai?

A: Chaar hypotheses test ki: timestamp misalignment (shift=0 hi sabse achha fit deta hai), inverter clipping (AC/DC ratio sabhi 22 inverters mein flat), sensor placement aur din-wise monsoon pattern. Root cause: shared weather sensor aur cloud variability, koi code bug nahi.

Q: Plant 1 itna achha kyun fit hua?

A: Plant 1 ka daytime r^2 = 0.97 confirm karta hai ki formula sound hai; Plant 2 ka weak fit isliye ek data limitation hai, formula ki galti nahi.

10 · Losses & PF

Tag: REAL (benchmark) + ILLUSTRATIVE (PF inputs)

Bolne wali script (~55 sec)

PF-correction kVAR aur I2R loss illustrative what-if hain, lekin saath mein real Delhi-DISCOM AT&C loss figures hain: BRPL 6.58%, BYPL 7.84%, TPDDL 5.91%, FY2023-24, seedhe PFC ke primary PDF se, Annexure 1.8, page 76. BRPL aur TPDDL secondary source se exact match karte hain; BYPL pehli baar confirm hua hai.

Kya hua / Kyu kiya / Kaise kiya

- Kya hua: Illustrative PF/current what-ifs, aur unke saath real AT&C benchmark.
- Kyu kiya: User apna PF-correction scenario try kar sake, saath mein real DISCOM numbers bhi dikhe.
- Kaise kiya: Losses ke AT&C numbers PFC ke primary PDF se seedhe nikale, secondary source se re-confirm kiye.

Formula / number ka matlab

$Qc = P(\tan\varphi 1 - \tan\varphi 2)$; $Ploss = 3I^2R$

Viva mein pooche ja sakte hain

Q: BYPL ka AT&C loss kitna hai?

A: 7.84%, FY2023-24, PFC ke primary PDF 'Report on Performance of Power Utilities 2023-24' (Annexure 1.8, page 76) se seedha. Isi table se BRPL 6.58% aur TPDDL 5.91% bhi.

Q: Pehle yeh BYPL ke bina kyun tha?

A: Secondary source (Power Line) mein BYPL genuinely mention nahi tha; is round mein primary PDF seedha dhoondh kar padha gaya.

Agli slide par: Ab aakhri do pages: theft detection aur ek summary data/model card.

Slide 12 — Theft & Segments, and the Data & Model Card

Slide par kya hai

Ek page mein dono features: Theft & Segments (real SGCC benchmark par LightGBM) aur Data & Model Card (saari 12 pages ka summary).

11 · Theft & Segments

Tag: FOREIGN-REAL (SGCC)

Bolne wali script (~60 sec)

> Humein Delhi ya kisi Indian DISCOM ka public consumer-level smart-meter dataset with theft labels nahi mila, isliye humne real SGCC dataset use kiya -- China State Grid ka public benchmark, Zheng et al. 2018 paper, 42,372 consumers, 8.53% theft rate. Logistic regression baseline se LightGBM tak gaye, test AUC-ROC 0.7585 mila, paper ke ~0.78 se thoda kam -- koi leakage red flag nahi, kyunki hamara number paper se neeche hai, upar nahi.

Kya hua / Kyu kiya / Kaise kiya

- Kya hua: Theft detection ek real foreign benchmark par.
- Kyu kiya: Jahan Delhi-specific data nahi mila, wahan honestly ek real (lekin foreign) benchmark use kiya.
- Kaise kiya: Logistic regression → LightGBM (Optuna-tuned) SGCC par; K-means segmentation; demo fleet (Isolation Forest) synthetic rakha.

Formula / number ka matlab

anomaly(i) = −IF.score(xi); cluster(i) = argmin_k $||xi - \mu k||^2$

Viva mein pooche ja sakte hain

Q: Theft ke liye SGCC (China) par kyun train kiya, Delhi data kyun nahi mila?

A: Humne explicitly search kiya ki koi public Delhi ya Indian DISCOM consumer-level smart-meter dataset with theft/anomaly labels maujood hai ya nahi -- nahi mila, isliye real foreign benchmark use kiya, fabricate nahi kiya.

Q: Demo fleet real hai?

A: Nahi, synthetic hai, sirf illustration ke liye jab tak real Delhi meter data available na ho.

12 · Data & Model Card

Tag: REAL (static)

Bolne wali script (~45 sec)

> Yeh ek static summary page hai -- har page ka data source, provenance aur result ek jagah, taaki koi bhi number ka trace mil sake. Live data nahi hai, docs/MODEL_RESULTS.md aur API responses se compile hota hai.

Kya hua / Kyu kiya / Kaise kiya

- Kya hua: Sab kuch ek summary page mein.

- Kyu kiya: Examiner ko ek hi jagah se sab pages' provenance dikhe.
- Kaise kiya: docs/MODEL_RESULTS.md aur API responses se compiled.

Formula / number ka matlab

coverage = (1/N) Σ 1[P10,i ≤ yi ≤ P90,i]

Viva mein pooche ja sakte hain

Q: Yeh page live data dikhata hai?

A: Nahi, static summary hai -- numbers docs se aate hain, per-request compute nahi hote.

Agli slide par: Yeh the humare saare 12 feature pages. Ab dekhte hain har page ka benchmark ek table mein.

Slide 13 — Benchmarks

Slide par kya hai

Table: har page ka benchmark, hamara result, aur source. Niche ek note jo is round ke Solar/Losses updates summarize karta hai.

Bolne wali script (~50 sec)

> Yeh slide har page ke result ko uske benchmark ke saamne rakhti hai, ek jagah par. Forecast: persistence 1.44/3.18% ke against humara test MAPE 3.93%, 84.4% coverage. IEX: EAC-PM ka range ₹1.11-9.71 ke against humara real history ₹3.05-6.13/unit. ToD: DERC clause 3(d) se seedha 20%/20%. Solar: do real plants se validate, r^2 0.99 aur 0.21 -- Plant 2 ka root cause ab investigate ho chuka hai. Losses: ab BRPL, BYPL aur TPDDL teeno PFC ke primary document se confirm hain. Theft: SGCC paper ke ~0.78 ke against humara 0.7585, no leakage.

Kya hua / Kyu kiya / Kaise kiya

- Kya hua: Saare benchmarks ek table mein.
- Kyu kiya: Examiner ko ek hi jagah dikhe ki kaunsa number measured hai, kaunsa external benchmark se compare hua.
- Kaise kiya: docs/BENCHMARKS.md se seedha liya, har row ka apna source.

Viva mein pooche ja sakte hain

Q: Sabse weak result kaunsa hai, aur kyun?

A: Solar Plant 2 (r^2 = 0.21) aur Theft (AUC 0.7585, paper se thoda kam). Dono ko humne khul kar report kiya, chhupaya nahi -- Plant 2 ka root cause investigate ho chuka hai, aur Theft ka number paper se neeche hai isliye leakage ka koi red flag nahi.

Agli slide par: Ab references aur phir closing.

Slide 14 — References

Slide par kya hai

20 sources: PIB, Delhi SLDC, CSE, EAC-PM, CERC, DERC, IEC 60076-7, textbooks (V.K. Mehta, Kothari & Nagrath, Gönen), Hyndman & Athanasopoulos (FPP3), conformal prediction, pandapower, Isolation Forest aur IEX market data.

Bolne wali script

> Slide par har external number aur equation ka source hai. Data ke liye Delhi SLDC aur Open-Meteo; regulation ke liye CERC DSM 2024 aur DERC; transformer ke liye IEC 60076-7; formulas ke liye V.K. Mehta aur Kothari & Nagrath; aur forecasting ke liye Hyndman ki FPP3 aur conformal prediction ka paper.

Kya hua / Kyu kiya / Kaise kiya

- Kya hua: Saare sources ek jagah.
- Kyu kiya: Har number verify kiya ja sake.
- Kaise kiya: Link aur textbook chapter ke saath.

Agli slide par: Aur ab, closing.

Slide 15 — Thank You

Slide par kya hai

Thank you, team aur guides ke naam.

Bolne wali script

> Toh sir/ma'am, humne ek real Delhi forecast ko uske nateejon tak le jaane ki koshish ki: peak alert, demand ki shape, garmi ka asar, DSM charge, transformer ageing, aur ab IEX cost, ToD billing, solar output, losses aur theft detection bhi usi forecast ke saath, har number par saaf label. Aapke time ke liye dhanyavaad. Ab hum aapke sawaalon ka swagat karte hain.

Kya hua / Kyu kiya / Kaise kiya

- Kya hua: Closing.
- Kyu kiya: Ek line mein project yaad dilana.
- Kaise kiya: Sawaalon ke liye invite.

Viva ke tough sawal

Har jawab ka number deck, report ya repo (BENCHMARKS.md, DATA_SOURCES_LOG.md, MODEL_RESULTS.md, DATA_QUALITY.md, engine code) se match karta hai.

1. Naive baseline kya hai aur kyun zaroori hai?

Seasonal-naive ka matlab: kal isi 15-minute block ka load hi aaj ka forecast. Isme koi learning nahi hai. Agar hamara model isse behtar nahi, to model ka koi matlab nahi.

2. LightGBM hi kyun? SARIMAX kyun drop hua?

LightGBM lags, weather aur calendar jaise bahut saare features ek saath le sakta hai, fast hai, aur SHAP se explain hota hai. SARIMAX (hourly) code mein hai lekin 15-minute, 96-block, multi-feature setup ke liye LightGBM zyada practical nikla.

3. Leakage kya hota hai? Hampel wala bug kaise pakda aur fix kiya?

Leakage matlab model ko training mein aisi jaankari mil jaana jo forecast ke time par hoti hi nahi, jaise future ka data. Pehle Hampel outlier filter future dekh raha tha; cleaning ko causal banaya gaya taaki har decision sirf past data par ho.

4. Naive ke 5.22% aur 5.16% mein farak kya hai?

Data same hai, rows alag hain. 5.16% poore test set par hai: woh 1,656,240 rows jinke liye kal ka value maujood hai. 5.22% model ladder table mein hai, jo thode alag subset par hai.

5. LF 0.646 aur 0.503 mein farak?

0.646 ek illustrative din hai: 530/820, sirf formula samjhane ke liye. 0.503 poore period (April 2023 - January 2026, 96,398 blocks) ka measured value hai.

6. P10-P90 ka target 80% hai aur coverage 84.4% hai. Iska matlab?

P10-P90 ek 80% range hai: theory mein 80% actual values iske andar aani chahiye. Test par 84.4% aayi, yaani band thoda conservative (chauda) hai. Yeh over-coverage hai, under-coverage nahi.

7. Test set fully untouched kyun nahi raha?

Pehle LightGBM ki test-set errors dekh kar (woh lambe horizons par naive se haar raha tha) humne target-aligned lag features design kiye. Hyperparameters validation par hi tune hue, lekin feature design ka decision test errors se influenced tha.

8. h = 1 par persistence model se behtar kyun hai?

15 minute mein load bahut kam badalta hai (lag-1 block autocorrelation 0.998). Isliye 'agla block = abhi wala block' h = 1 par 1.44% MAPE deta hai, jabki LightGBM 3.18%.

9. Diwali par error zyada kyun hai?

Diwali (20 October 2025) par day MAPE 8.65% tha aur sirf 49% blocks band mein aaye. Actual load 4-week average se lagbhag 2,000 MW kam tha. Model holiday flag jaanta hai lekin Diwali-specific pattern ko poori tarah capture nahi karta.

10. Synthetic aur illustrative data kahan use hua aur kyun?

Synthetic: paanch consumer category profiles (residential, commercial, industrial, agricultural, EV), parametric model se bane, ab sirf Theft & Segments ke demo fleet ke liye. Illustrative: DSM

₹ Calculator, Transformer Health, aur ToD Bill Simulator ka consumer load profile -- formula real hai, input hypothetical.

11. DSM ₹ figure illustrative kyun hai?

CERC DSM Regulations 2024 mein normal rate grid frequency par depend karta hai. Woh tables encode nahi kiye; humne flat ₹5,000/MWh aur k = 1.5 maana. Isliye number directional hai, exact regulatory penalty nahi.

12. 8,748 MW wala peak dataset mein kyun nahi?

Dataset 12 January 2026 par khatam hota hai, aur yeh record 29 June 2026 ka hai. Data ka maximum 8,581.6 MW (19 June 2024) hai. Isliye test window mein yeh peak maujood nahi.

13. Overfitting hai ya nahi?

Train, validation aur test MAPE 2.29%, 4.29% aur 3.93% hain (accuracy 97.71%, 95.71%, 96.07%). Gap chhota hai: early stopping ne 3,000 mein se 436 rounds par roka, aur validation/test error flat rahi. Isliye overfitting nuksaan wali nahi hai.

14. Data clean karne mein kitna badla gaya?

15-minute ke 97,728 blocks mein se 2,047 fill kiye gaye (894 forward-fill, 1,153 profile-fill), 249 outlier flag hue, aur 1,330 blocks khaali chhode. Sab decisions causal hain.

15. 96% accuracy kaise nikali?

Accuracy = 100 − MAPE. Test MAPE 3.93% hai, isliye accuracy 96.07% (≈ 96%); train par 97.71%, validation par 95.71%. Yeh classification accuracy nahi hai.

16. Solar Plant 2 ka r^2 itna kam (0.21) kyun hai?

Humne chaar hypotheses test ki: timestamp misalignment (shift karke dekha, par shift=0 hi sabse achha fit deta hai), inverter clipping (AC/DC ratio sabhi 22 inverters mein flat), sensor placement aur din-wise monsoon pattern. Root cause: shared weather sensor aur cloud variability, koi code bug nahi.

17. BYPL ka AT&C loss kitna hai, aur kaise pata chala?

BYPL ka AT&C loss 7.84% hai, FY2023-24. Pehle yeh secondary source (Power Line) mein nahi mila tha, isliye unconfirmed rakha tha. Is round mein humne PFC ka primary PDF seedha dhoondh kar padha.

18. Theft & Segments SGCC (China) par kyun train kiya, Delhi data kyun nahi mila?

Humne explicitly search kiya ki koi public Delhi ya Indian DISCOM consumer-level smart-meter dataset with theft/anomaly labels maujood hai ya nahi -- nahi mila, isliye real foreign benchmark use kiya.

19. IEX price ka number kaise verify kiya?

Humara number Kaggle ke real IEX day-ahead market history (2018-2024, 233,760 rows) ka block-of-day average hai. Isko IEX ki apni published price range ke saath compare kiya, match karta hai.